

STUDENT PERFORMANCE DASHBOARD REPORT

1. ABSTRACT

This project focuses on analyzing student performance data using Python and Power BI to derive meaningful academic insights. The dataset includes information about student marks, subjects, attendance, grades, and overall performance.

The main objective of this project is to evaluate student performance, identify strengths and weaknesses, and improve academic outcomes. The project involves data preprocessing using Python libraries such as Pandas, followed by the creation of an interactive Power BI dashboard.

The dashboard highlights key metrics such as average marks, pass percentage, attendance rate, and subject-wise performance. Various visualizations like bar charts, pie charts, and line charts are used to present insights effectively.

2. INTRODUCTION

In the education sector, analyzing student performance is essential for improving teaching strategies and academic results. Educational institutions generate large amounts of data related to student marks, attendance, and assessments.

This project aims to analyze student data and create a dashboard that provides a complete overview of academic performance. Python is used for data cleaning and preprocessing, while Power BI is used for creating interactive visualizations.

The dashboard helps teachers and administrators track student progress and make data-driven decisions.

3. OBJECTIVES

- To clean and preprocess student dataset
 - To analyze student performance across subjects
 - To identify top-performing and low-performing students
 - To evaluate attendance impact on performance
 - To create an interactive Power BI dashboard
 - To generate insights for academic improvement
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4. DATA PREPROCESSING

The dataset is processed using Python (Pandas) with the following steps:

- Removal of duplicate records
- Handling missing values
- Formatting subject-wise marks
- Creating new metrics like:
 - **Total Marks = Sum of all subjects**
 - **Percentage = (Total Marks / Maximum Marks) × 100**

These steps ensure clean and structured data for analysis.

5. EXPLORATORY DATA ANALYSIS

5.1 Subject-wise Performance

Analyzes marks distribution across different subjects

5.2 Top Students

Identifies students with highest percentage

5.3 Attendance Analysis

Examines relationship between attendance and performance

5.4 Grade Distribution

Analyzes how students are distributed across grades

These analyses help understand academic trends and student behavior.

6. POWER BI DASHBOARD

6.1 Key Performance Indicators (KPIs)

- Average Marks
- Pass Percentage
- Total Students
- Average Attendance

6.2 Visualizations Used

- **Bar Chart:** Subject-wise average marks
 - **Pie/Donut Chart:** Grade distribution
 - **Line Chart:** Performance trend over time
 - **Column Chart:** Attendance vs marks
 - **Table:** Student-wise performance
 - **Slicers:** Class, subject, and grade filters
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7. KEY INSIGHTS

- Students perform differently across subjects
 - Higher attendance leads to better performance
 - Few students achieve top grades consistently
 - Some subjects require improvement focus
 - Performance trends help identify weak areas
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8. ADVANTAGES

- Easy visualization of academic data
 - Interactive dashboard
 - Helps teachers monitor performance
 - Supports data-driven decisions
 - Improves student outcomes
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9. LIMITATIONS

- Limited real-time data
 - No predictive analytics
 - Depends on data quality
 - Limited behavioral insights
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10. FUTURE ENHANCEMENTS

- Add real-time student data integration
 - Implement performance prediction using ML
 - Add student feedback analysis
 - Improve dashboard interactivity
 - Deploy on web platform
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11. CONCLUSION

The Student Performance Dashboard successfully demonstrates how Python and Power BI can be used to analyze academic data effectively.

The dashboard provides valuable insights into student performance, attendance, and subject-wise trends, helping educators improve teaching strategies. With further enhancements, it can become a complete academic analytics system.